

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and iterated on **rapid prototypes** for novel robotics concepts, evaluating system feasibility and contributing to architectural design documentation at HCL Technologies.
- Engineered **Python automation scripts** for data processing and ETL tasks, improving data pipeline efficiency for internal operational reporting at HCL Technologies.
- Designed and implemented **Java Spring Boot REST APIs** for cross-service communication, enhancing system integration capabilities within enterprise applications at HCL Technologies.
- Built and maintained **end-to-end platform architectures** and reference implementations for robotics systems, integrating sensing, compute, connectivity, and motion subsystems.
- Executed **characterization and benchmarking tests** for robotics subsystems, generating performance and robustness data to inform architecture choices and technology readiness.
- Produced **design documentation and integration guides** for robotics platforms, capturing system tradeoffs and recommended patterns for transfer to product teams.
- Containerized backend services using **Docker** and deployed them to **AWS EC2 instances**, configuring basic resource limits for predictable rollouts.

SKILLS

Robotics Software & Controls – ROS/ROS2, Scripting, Data Analysis, System Integration, Embedded Firmware, Real-time Development, Control Loops, Drivers, Timing, Interfaces

Hardware & Electronics – Lab Measurement, Hardware Bring-up, Sensing Subsystems, Power Subsystems, Motion Subsystems

Software Development – Java, Spring Boot, Python, REST APIs, Docker, Linux, Git, Jenkins, Jira, Confluence

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Kubernetes

Data & Analytics – SQL, PostgreSQL, MySQL, Oracle, Power BI, Tableau, R/RStudio, pandas, NumPy, scikit-learn, DAX, Power Query, Data Modelling

Enterprise Systems – ERP Integration, Change Management, Payments Operations Dashboard, Transaction Monitoring, Reconciliation Workflows

PROJECTS

Robotics System Integration Framework (HCL Technologies)

- Developed a modular integration framework in **Java** for connecting diverse robotics subsystems, including sensing, motion control, and data telemetry.
- Implemented **REST APIs** for inter-component communication and data exchange, establishing clear contracts for system interactions.
- Containerized core services using **Docker**, enabling consistent deployment across development and testing environments.

Automated Robotics Test Suite Development (MSc Project)

- Designed and implemented an automated test suite using **Python** to validate the performance and robustness of robotic motion control algorithms.
- Integrated **lab measurement** tools and simulated sensor data to generate realistic test conditions for characterization.
- Analyzed test results using **pandas** and generated reports to identify performance bottlenecks and areas for improvement.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)